Student Seminar - IDP Presentation 5

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Tests

Testset: 2873 scraped papers with XML full-text available

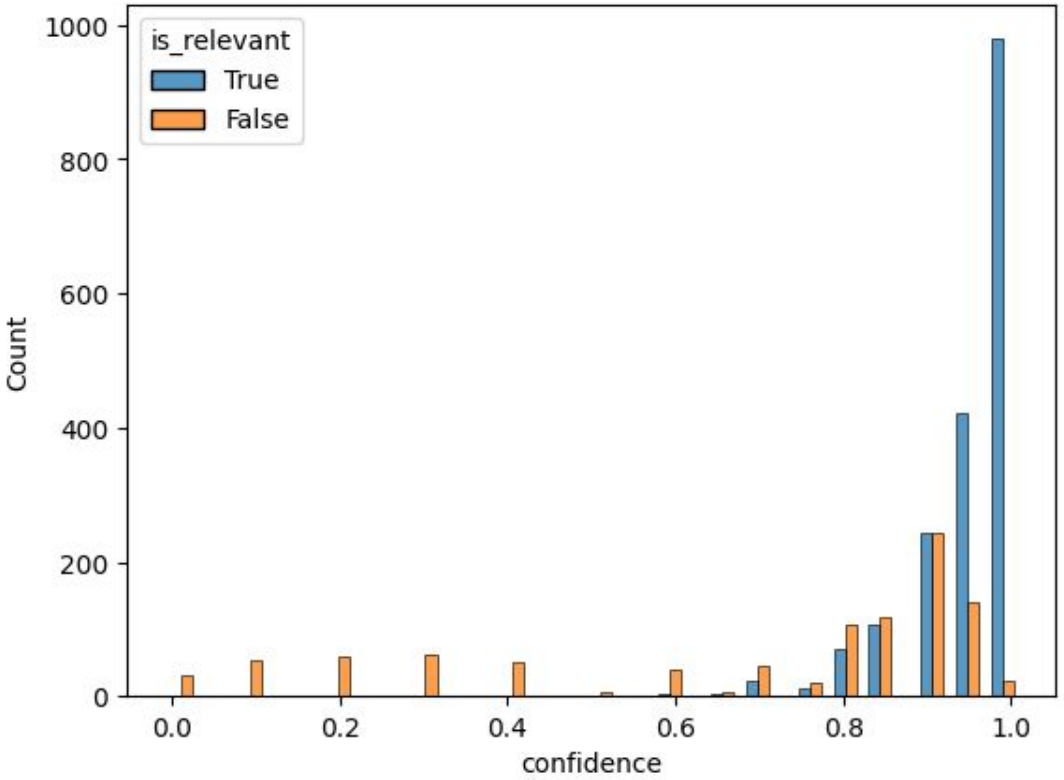
Relevance Check (gemini-3.5-flash-lite)

Extraction (gemini-3.5-flash-lite)

Evaluation (mistral-small3.2:24b)

Relevance Results

is_relevant (2873)
- **True: 1869**
- **False: 1004**

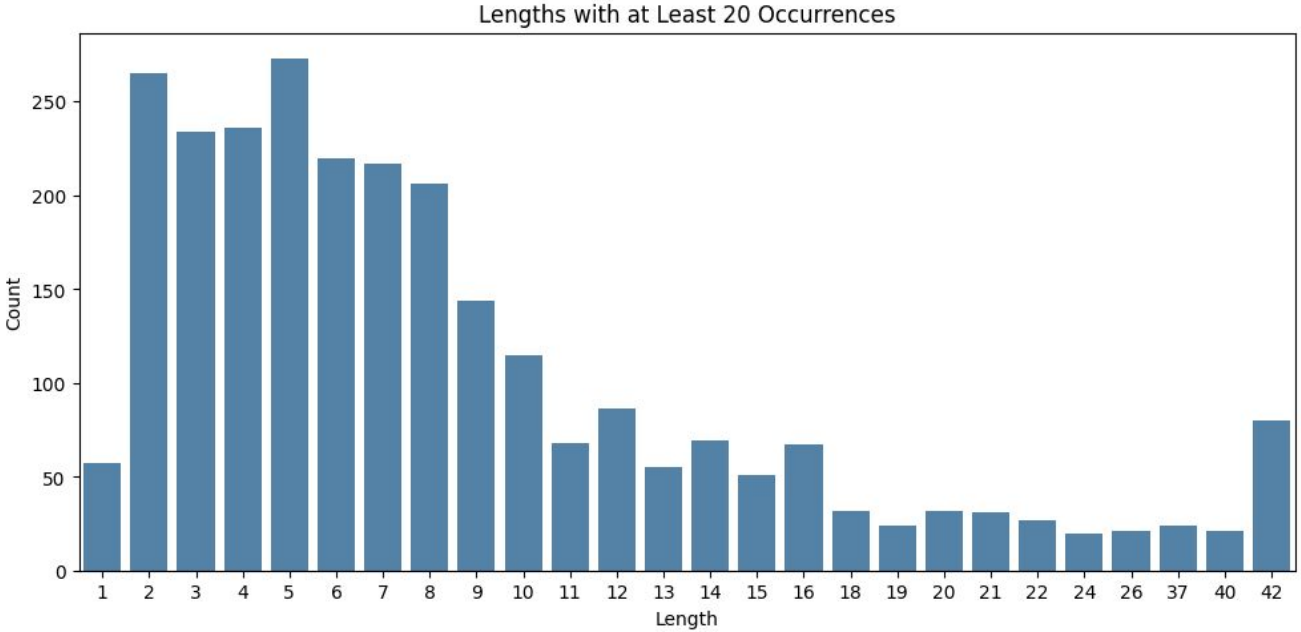


Extraction Results

Used 1490/1869 relevant papers

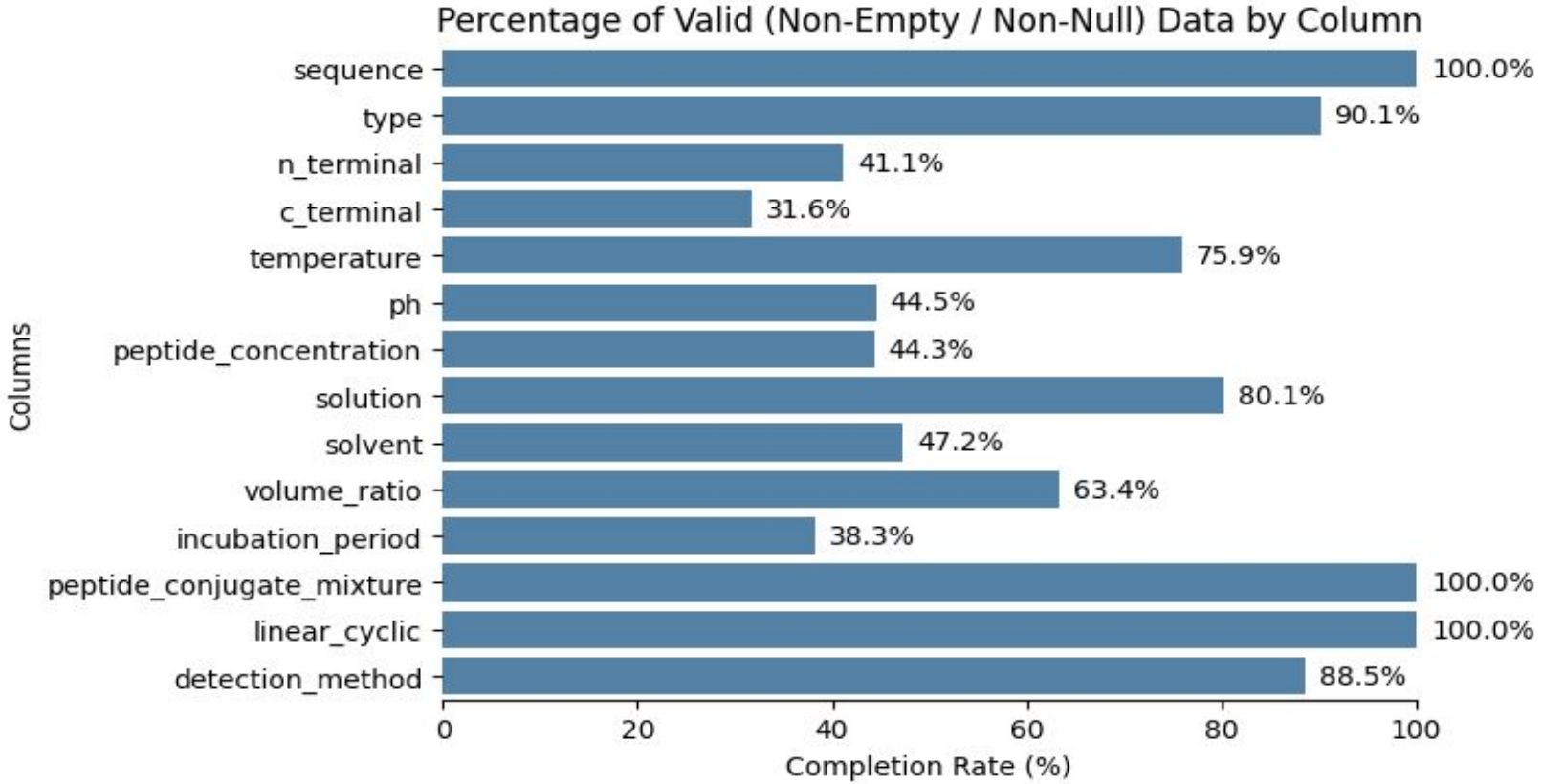
38 papers had no entries

2866 total entries

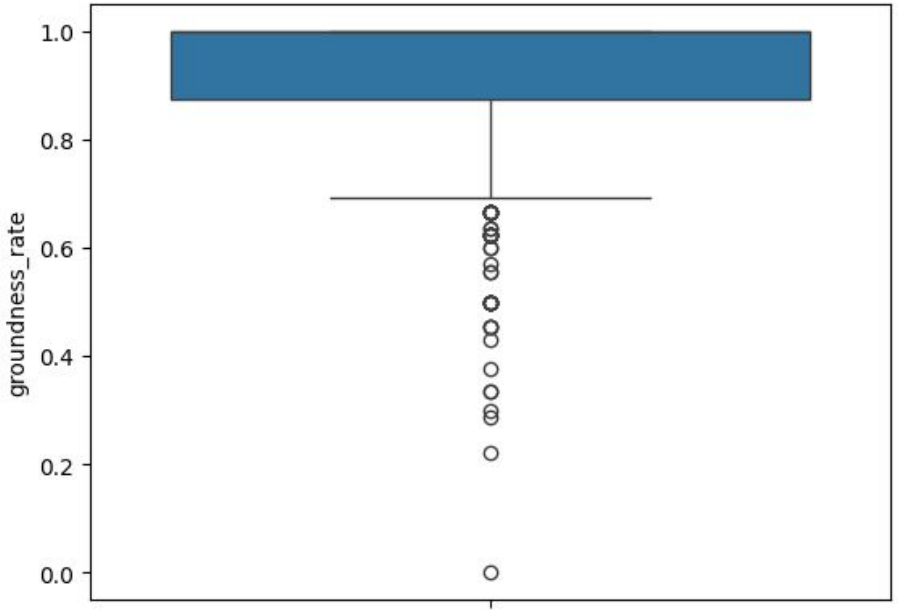


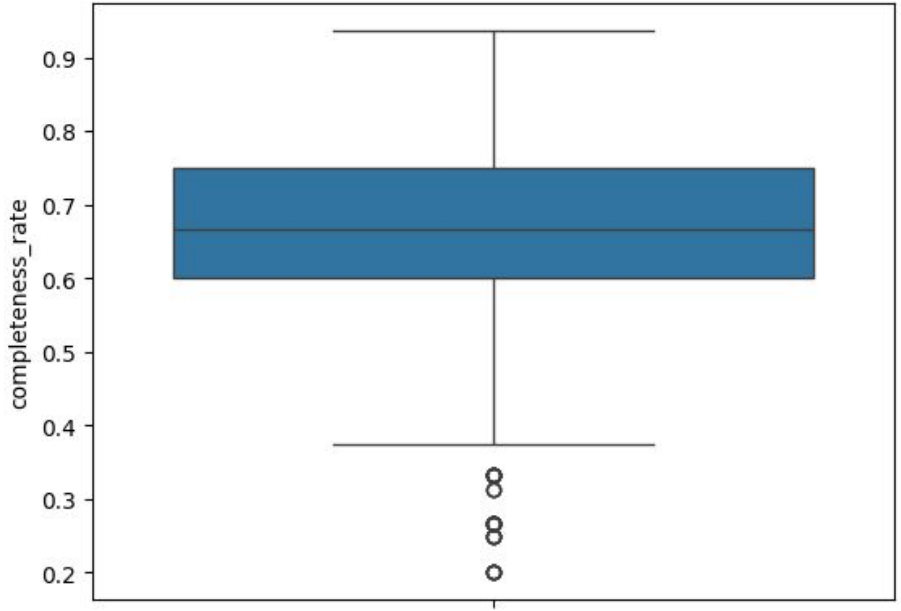
- Papers with no extracted entries -> Potential improvement to relevance check

Extraction Results (2)



Evaluation Results (only 500/2866 extraction entries)





Considerations (1)

- Google API is much faster than GPU server (low priority):
  Relevance Check Task (batches of 10 for 40 papers):
  - gemini-3.5-flash-lite: 0.6 seconds/paper
  - mistral-3.2small: 6.25 seconds/paper

Relevance check for 2873 papers took <30 minutes with google api vs ~20 hours (estimate) on gpu server

Considerations (2)

- Small models like mistral-small3.2 (24b) / gemma 4 (31b) perform really poorly on extraction task:

1)


```
{ "entries": [
 { "sequence": "FKFEFKFE", "evidence_span":
{ "sequence": "[sec-0/p-0/s-3]" } },
 { "sequence": "FKFEFKFE", "evidence_span":
{ "sequence": "[sec-0/p-0/s-5]" } },
 { "sequence": "FKFEFKFE", "evidence_span":
{ "sequence": "[sec-0/p-3/s-0]" } },
 { "sequence": "FKFEFKFE", "evidence_span":
{ "sequence": "[sec-0/p-3/s-1]" } },
 { "sequence": "FKFEFKFE", "evidence_span":
{ "sequence": "[sec-0/p-3/s-2]" }
 ...
```


2)


```
{ "entries": [
 {
 "sequence": "FKFEFKFE",
 "type": "amyloid",
 "ph": 7,
 "conjugate_partner": "N-acetylated and C-amidated",
 "preparation_protocol": "....",
 "evidence_span": {
 "type": "paragraph",
 "solvent": "water",
 "conjugate_partner": "N-acetylated and C-amidated"
 }
 },
 {
 "sequence": "FKFEFKFE",
 "type": "amyloid",
 "ph": 7,
 "conjugate_partner": "N-acetylated and C-amidated",
 "preparation_protocol": "...",
 "evidence_span": {
 "type": "paragraph",
 "solvent": "water",
 "conjugate_partner": "N-acetylated and C-amidated"
 }
 }
```

'**sequence**': ['FKFEFKFE'],
'**type**': ['ribbon'],
'**n_terminal**': [None],
'**c_terminal**': [None],
'**temperature**': [25],
'**ph**': [None],
'**peptide_concentration**': [None],
'**solution**': ['water'],
'**solvent**': [None],
'**volume_ratio**': ['1'],
'**incubation_period**': [None],
'**peptide_conjugate_mixture**': ['peptide'],
'**conjugate_partner**': [None],
'**linear_cyclic**': ['linear'],
'**detection_method**': ['Cryo-EM'],
'**preparation_protocol**': ['The Cryo-EM structure of the FKFEFKFE peptide (PDB ID: 7LQF) was downloaded.'],
'**evidence_span**':
['{"sequence":"[sec-0/p-0/s-3]","type":"[sec-0/p-0/s-3]","temperature":"[sec-1/sec-2/p-1/s-1]","solution":"[sec-0/p-1/s-0]"}'],

Questions

1) For extraction, should we stick with large models (like Gemini 3.5 Flash Lite) or adapt the code to make it work with small models?

Notes

PMC Open Access Subset returned 55k full-text XML for the query: “**peptide self-assembly**”. Next step could be to use these papers (relevance -> fetch -> process)